

# Case Study

## Fraud Detection for Financial Services

Used Technologies: Artificial Intelligence (AI), Machine Learning (ML), Anomaly Detection, Python, PyTorch, Scikit-learn, XGBoost, FastAPI, PostgreSQL, REST APIs, React, TailwindCSS, Docker, Azure Machine Learning

### Executive Summary

Robles.AI partnered with a leading financial institution in Latin America to implement an AI system capable of detecting fraudulent transactions in real-time.

Within just 4 months of deployment, the system achieved 99.2% accuracy in fraud detection and helped the institution save more than \$4.5 million annually in fraud-related losses.

The success of this project has become a benchmark for the application of AI in the regional financial sector.



### System Components and Architecture

- **Data Source:** Transaction logs, user behavior history, geolocation, date and time, device fingerprinting, and historical fraud patterns.
- **AI Engine:** Ensemble ML models using XGBoost and neural networks trained on more than 6 million labeled transactions.
- **Feature Engineering:** Real-time feature generation (time windows, amount deviation, country mismatches) implemented in Python and PyTorch.
- **Backend and APIs:** Built with FastAPI, exposing REST endpoints for authorization and risk scoring.
- **Real-Time Infrastructure:** Deployed in Docker containers orchestrated on Azure Kubernetes Service (AKS) for scalable low-latency processing.
- **Dashboard:** React-based admin panel showing fraud heatmaps, alerts, and user behavior trends.

### Project Stages

1. **Data Analysis and Preparation**
  - Anonymized transactional records and enriched with behavioral features.
  - Domain experts labeled 250,000 transactions to train the first model.
2. **Model Training and Testing**
  - Cross-validation and hyperparameter tuning performed on Azure ML.
  - Achieved 99.2% precision, 97% recall.
  - Simulated in a shadow environment for 2 weeks before going live.
3. **Integration and Deployment**
  - Integrated with the banking system's transaction processor.
  - Risk scores trigger automated responses or alert a fraud analyst.
4. **Monitoring and Continuous Learning**
  - Retrained monthly with new cases.
  - Feedback loop from fraud analysts improves future detection.

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### FRAUD DETECTION FOR FINANCIAL SERVICES



*"Thanks to Robles.AI, we've turned fraud prevention into a competitive advantage. The system not only detects threats, but also empowers our analysts with better insights. We're setting new industry standards."*

— Luis Ramírez, Head of Risk Management  
Financia LATAM

#### Results Achieved

- **99.2% fraud detection accuracy.**
- **\$4.5M+ in annual savings** by preventing fraudulent transactions.
- **Reduction in false positives by 36%**, improving customer experience.
- **Response time under 300ms**, enabling real-time decisions.

#### Business Impact

The implementation enabled the bank to act proactively against fraud attempts and reduce operational burdens on investigation teams. The high precision of the model minimized disruptions for legitimate customers, while the integration with internal dashboards enhanced analyst decision-making.

Executives valued the transparency of the AI engine, especially the explainability module based on SHAP values. It allowed regulatory compliance and increased trust in automated systems.

#### Conclusion

This case study demonstrates how tailored AI systems can dramatically improve fraud detection capabilities for financial institutions. Robles.AI remains committed to enhancing trust, agility, and resilience in the financial sector through state-of-the-art technology.